

College Job & Internship Portal

Day 2 – Design Document

Presented by: Avika Bindal

Date: June 18, 2026

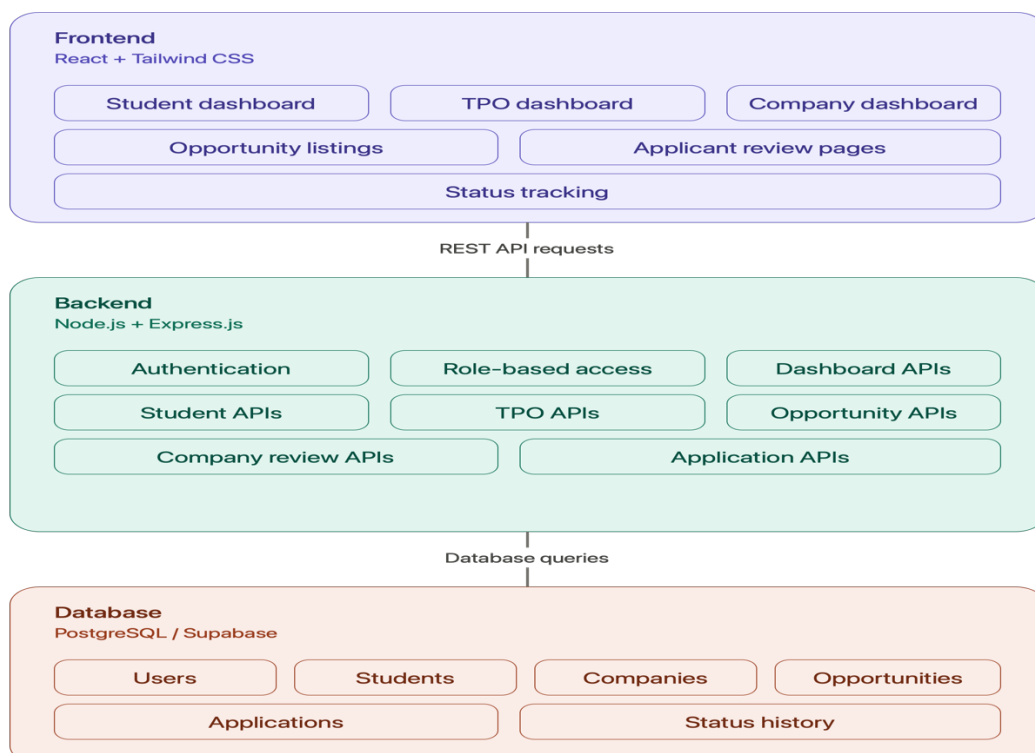
1. PROJECT OVERVIEW

The Campus Placement Portal is a full-stack web application designed to manage college placement and internship activities in one centralized system.

In this system, the Training and Placement Officer (TPO) posts job and internship opportunities on behalf of companies. Students can register, create their profiles, view available opportunities, and apply through the portal. Companies do not post opportunities directly; instead, they can view applicants for their opportunities and shortlist suitable students.

The TPO manages the complete placement workflow, including company details, opportunity posting, student applications, and application status tracking. This helps reduce manual communication through WhatsApp, emails, and spreadsheets, and provides a clear dashboard for students, companies, and the TPO.

2. ARCHITECTURE DIAGRAM



3. ER DIAGRAM

ER Relationships

USERS 1 ----- 1 STUDENTS

USERS 1 ----- 1 TPO

USERS 1 ----- 1 COMPANIES

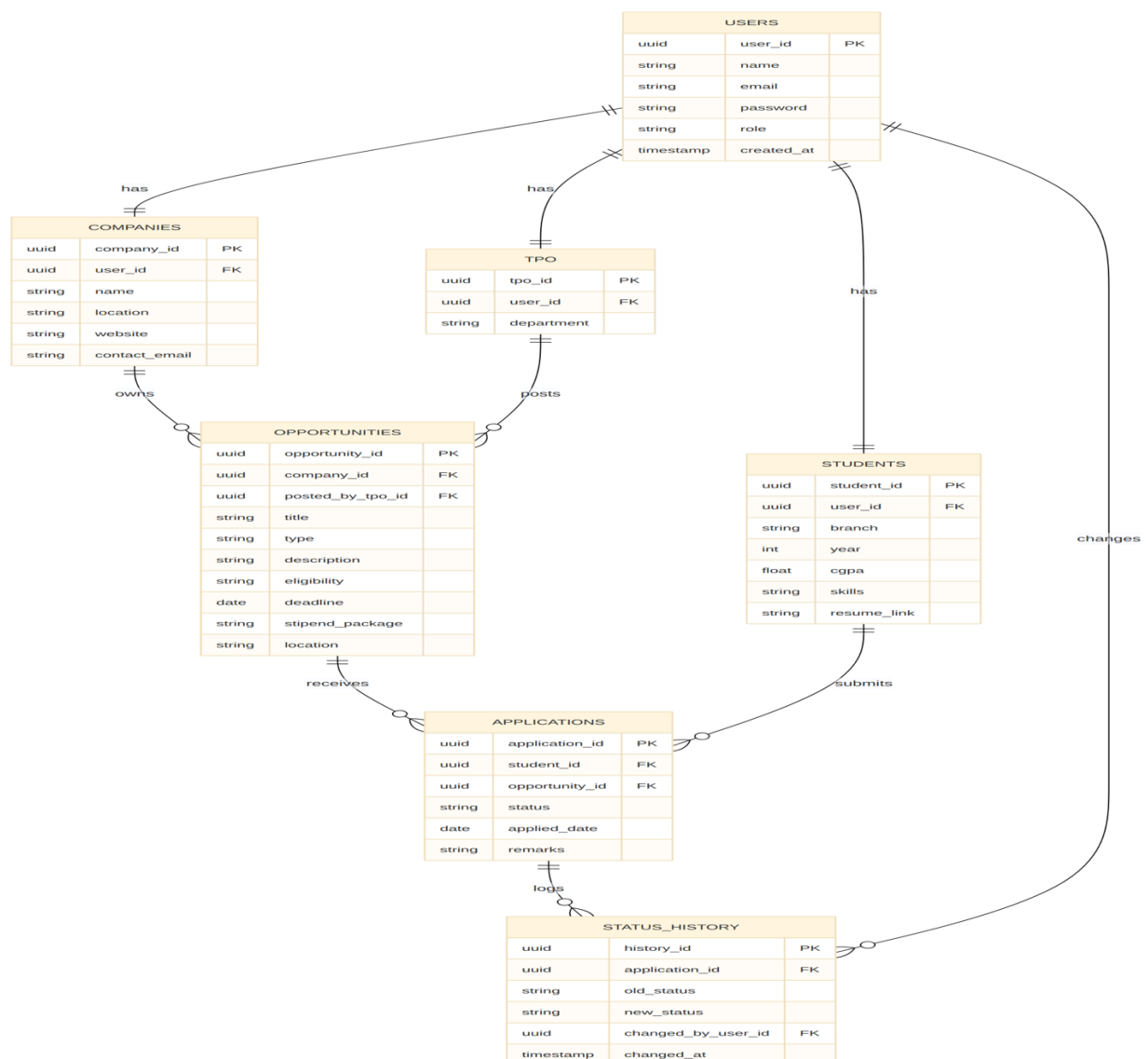
TPO 1 ----- many OPPORTUNITIES

COMPANIES 1 ----- many OPPORTUNITIES

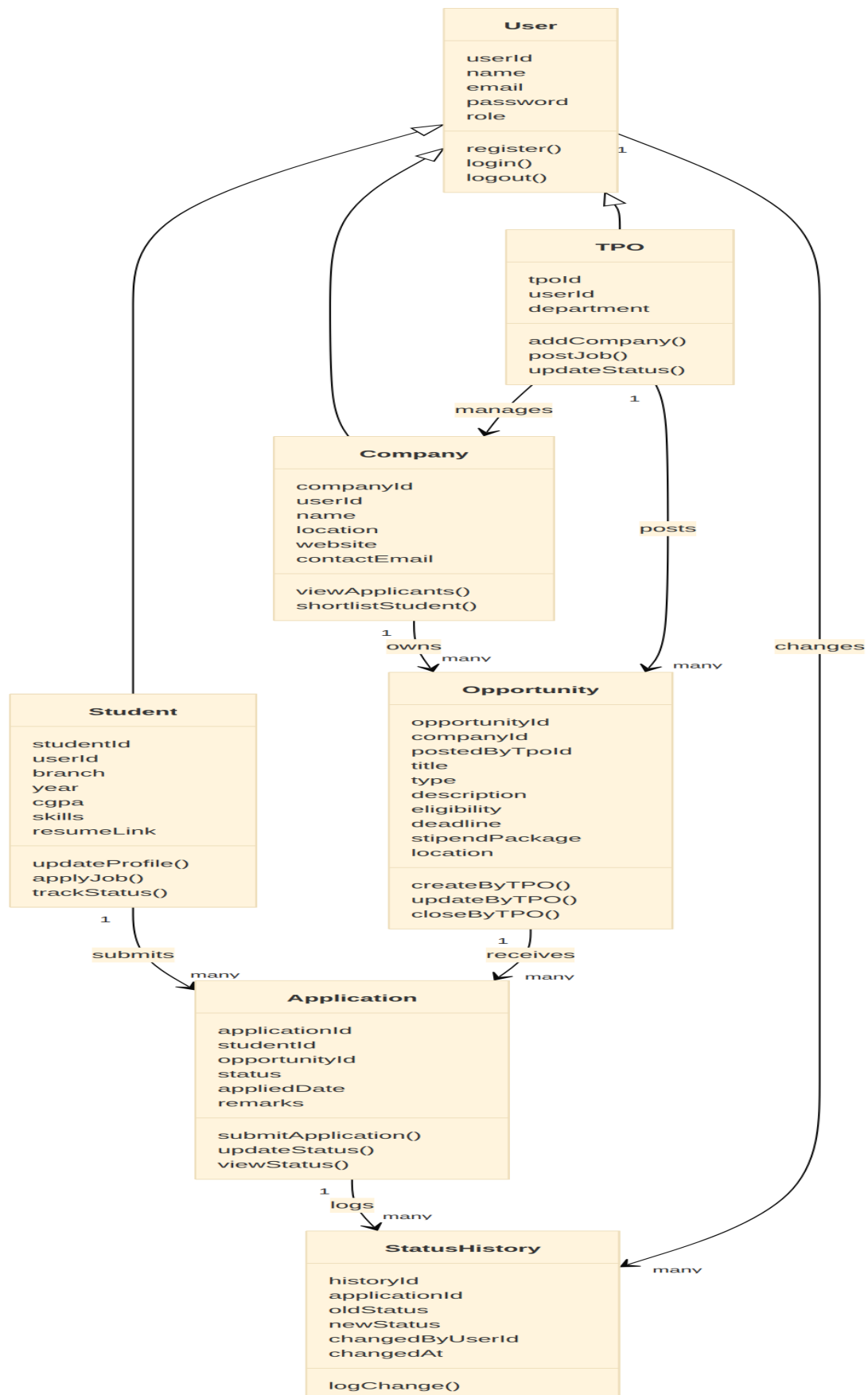
STUDENTS 1 ----- many APPLICATIONS

OPPORTUNITIES 1 ----- many APPLICATIONS

STUDENTS many ----- many OPPORTUNITIES through APPLICATIONS



4. CLASS DIAGRAM



5. API LIST

Endpoint	Method	Description	Access
/users/register	POST	Register a new user (student, tpo, or company)	Public
/users/login	POST	Login user	Public
/users/logout	POST	Logout user	All (authenticated)
/users/me	GET	Get logged-in user details	All (authenticated)
/students/profile	GET	Get student profile	Student
/students/profile	POST	Create student profile	Student
/students/profile	PUT	Update student profile	Student
/companies	GET	Get all companies	TPO
/companies	POST	Add new company (creates linked user account)	TPO
/companies/:id	GET	Get company by ID	TPO, Company (own)
/companies/:id	PUT	Update company profile	TPO, Company (own)
/companies/:id	DELETE	Delete company	TPO
/opportunities	GET	Get all opportunities	Student, TPO, Company (own)
/opportunities	POST	Create new opportunity	TPO
/opportunities/:id	GET	Get opportunity details	Student, TPO, Company (own)
/opportunities/:id	PUT	Update opportunity	TPO
/opportunities/:id	DELETE	Close/delete opportunity	TPO
/applications	GET	Get applications	TPO
/applications	POST	Apply for an opportunity	Student
/applications/student/:studentId	GET	Get applications of a student	Student (own), TPO
/applications/opportunity/:id	GET	Get applicants for an opportunity	TPO, Company (own opportunity)
/applications/:id/status	PUT	Update application status (shortlist/reject)	TPO, Company (own opportunity)
/dashboard/tpo	GET	Get TPO dashboard data	TPO
/dashboard/student	GET	Get student dashboard data	Student
/dashboard/company	GET	Get company dashboard data	Company

6. FUTURE SCOPE

If given 6 months and a budget, the project can be expanded with:

1. Email and SMS notifications

Send automatic alerts for new opportunities, deadlines, shortlisting, assessments, and interviews.

2. Resume upload and storage

Allow students to upload resumes directly instead of only adding resume links.

3. Resume parsing

Extract skills, education, experience, and contact details automatically from uploaded resumes.

4. AI-based student matching

Recommend suitable students to companies based on skills, CGPA, branch, year, and opportunity requirements.

5. Advanced company review tools

Allow companies to add review comments, shortlist students, reject applicants, and share feedback with the TPO.

6. Interview scheduling

Allow TPO and companies to schedule interviews and assessment rounds inside the portal.

7. Calendar integration

Sync interview dates and deadlines with Google Calendar or Outlook Calendar.

8. Bulk student import

Allow TPO to upload student data using CSV or Excel files.

9. Advanced analytics dashboard

Show department-wise applications, selection ratio, company-wise hiring, and placement statistics.

10. Offer letter management

Allow TPO to upload and manage offer letters and final placement records.

11. Multi-college or multi-department support

Support different departments, batches, and placement coordinators.

12. Mobile app

Create a mobile app so students can quickly view opportunities and track application status.